

A TECHNICAL SEMINAR PRESENTATION REPORT

on

DECEPTION TECHNOLOGY IN CYBER SECURITY

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD

In partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

By

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Under the Guidance of
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NAAC Accredited Institution with 'A' Grade & Recognized Under Section 2(f) & 12(B) of the UGC act, 1956

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(CYBER SECURITY)

MARCH, 2026



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CERTIFICATE

This is to certify that the technical seminar presentation report entitled “**DECEPTION TECHNOLOGY IN CYBER SECURITY**” work done by **KONGA BHAVANI [227Y1A6209]** student of Department of Computer Science and Engineering (Cyber Security), is a record of bonafide work carried out by the member during a period from December 2025 to March 2026 under the supervision of Mr. Gunna Manoj, Assistant Professor. This project is done as a fulfilment of obtaining Bachelor of Technology Degree to be awarded by Jawaharlal Nehru Technological University Hyderabad, Hyderabad.

The matter embodied in this technical seminar presentation report has not been submitted by me to any other university for the award of any other degree.

KONGA BHAVANI [227Y1A6209]

This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Date:

(Mr. Gunna Manoj)

The technical seminar presentation by the above student was held on

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DECLARATION

I hereby declare that the technical seminar presentation report entitled “DECEPTION TECHNOLOGY IN CYBER SECURITY” is bonafide work duly completed by me. It does not contain any part of the report or thesis submitted by any other candidate to this or any other institute of the university.

All such materials that have been obtained from other sources have been duly acknowledged.

KONGA BHAVANI [227Y1A6209]

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ABSTRACT

Deception technology is an advanced cybersecurity approach that proactively detects, misleads, and analyzes cyber threats by deploying decoys, traps, and fake assets within a network environment. Unlike traditional security mechanisms that focus on prevention and perimeter defense, deception technology operates on the principle of creating a controlled environment where attackers are lured into interacting with fabricated systems, credentials, or data. This enables early threat detection, real-time monitoring, and detailed attack analysis without alerting malicious actors.

Modern cybersecurity frameworks incorporate deception solutions such as honeypots, honeytokens, and decoy servers to identify insider threats, advanced persistent threats (APTs), and zero-day exploits. Platforms like Illusive Networks, TrapX Security, and Attivo Networks provide enterprise-grade deception systems that integrate with existing security infrastructures. These solutions enhance visibility across endpoints, networks, and cloud environments while reducing dwell time and false positives.

The benefits of deception technology extend across industries such as finance, healthcare, government, and critical infrastructure. By diverting attackers away from critical assets, organizations can reduce data breaches, strengthen incident response, and gather actionable threat intelligence. Deception mechanisms also improve compliance and risk management strategies by identifying vulnerabilities before exploitation.

Despite its advantages, deception technology faces challenges including deployment complexity, maintenance overhead, and evolving attacker awareness. Ensuring realistic decoy environments and seamless integration with Security Operations Centers (SOCs) remains critical. As cyber threats continue to grow in sophistication, advancements in artificial intelligence and automated response systems are further enhancing the effectiveness and scalability of deception-based defense strategies.